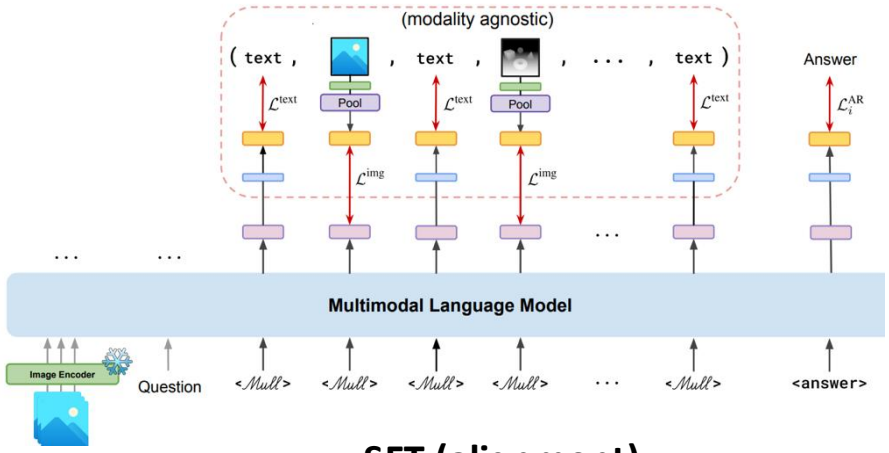
Latent reasoning for spatial reasoning

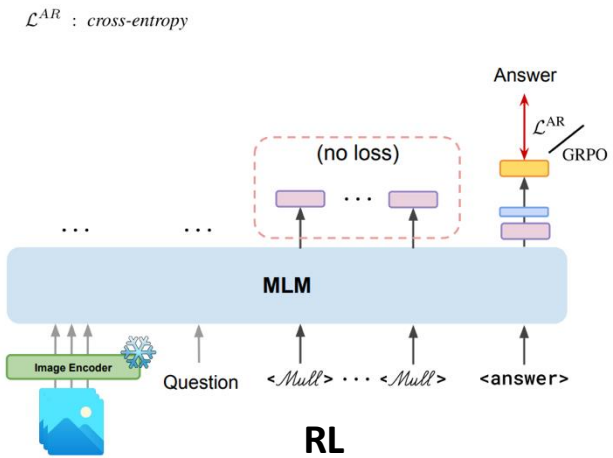
Three potential pathways

1 latent alignment + RL

1 teach the model the generate token in a semantically rich latent space (e.g. learn to generate VGGT embeddings)
2 does not teach the model how to leverage the generated latent information for reasoning; instead, the reasoning capability relies entirely on reinforcement learning



SFT (alignment)

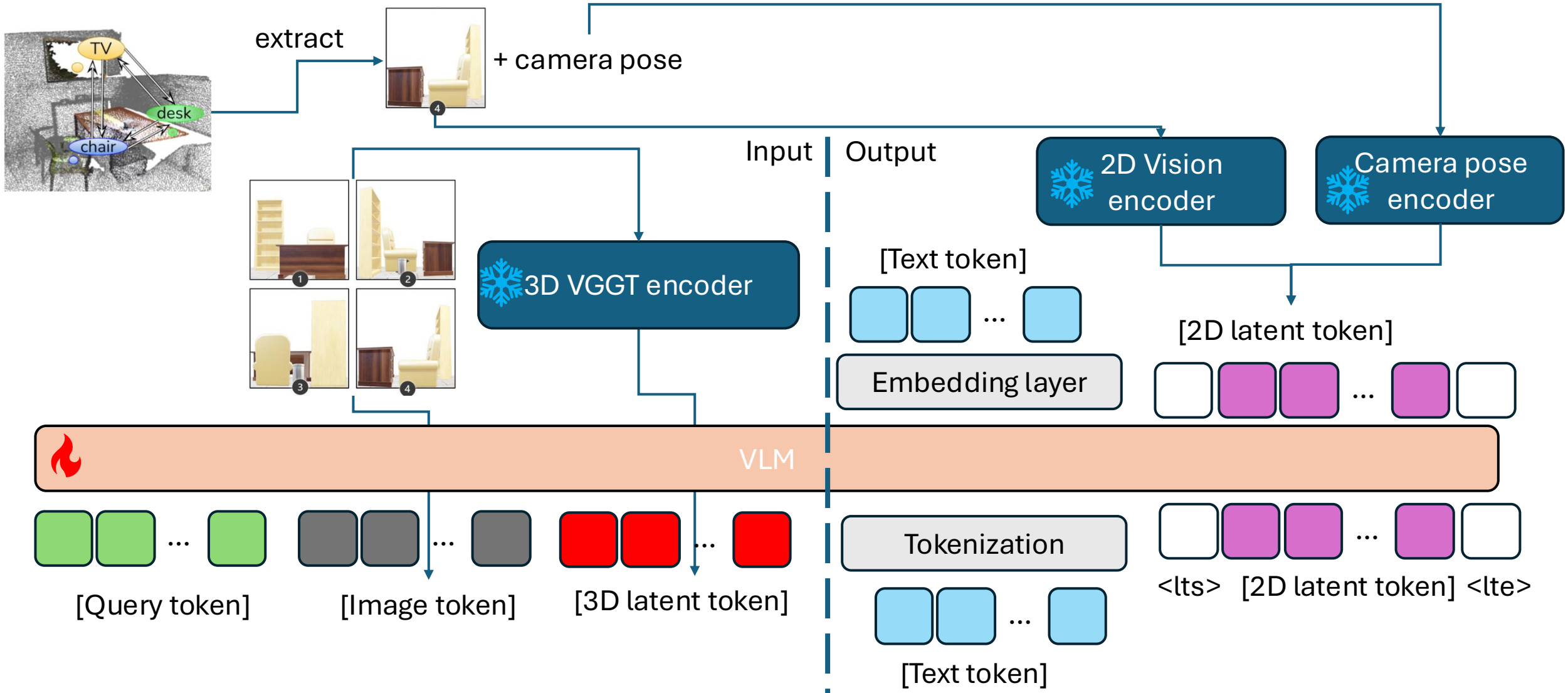


RL

Potential idea: Latent spatial reasoning

❖ Latent spatial reasoning potential idea

Extract step-level spatial cues to supervise reasoning during view shifts.

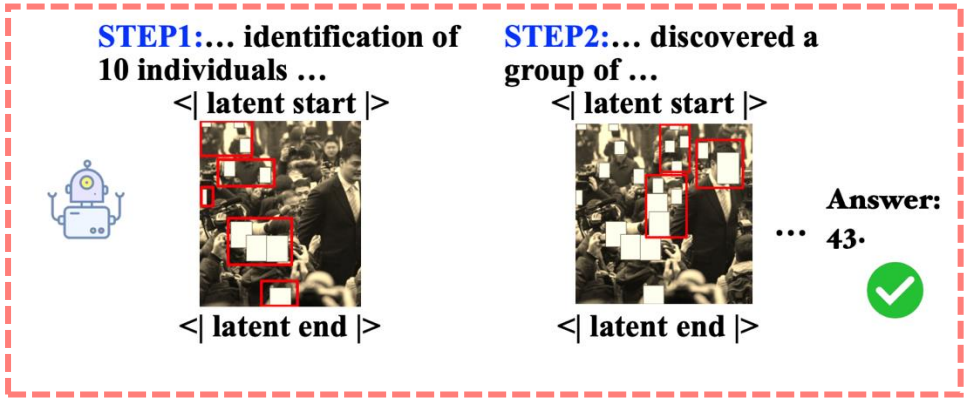


Latent reasoning for spatial reasoning

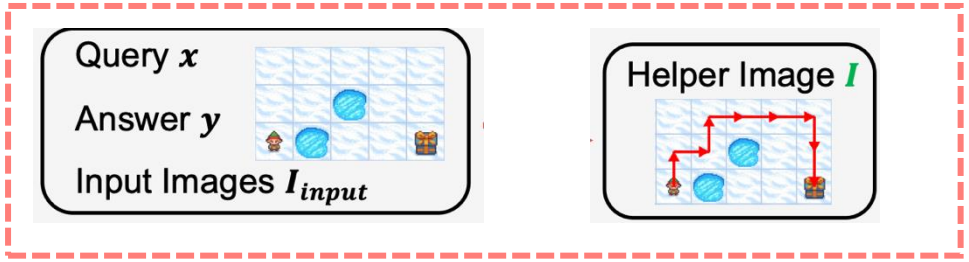
Three potential pathways

2 guide the reasoning with ground truth reasoning traces

Option 1: teach the model to focus on the correct part of input



Option 2: teach the model with mental imaginary operation

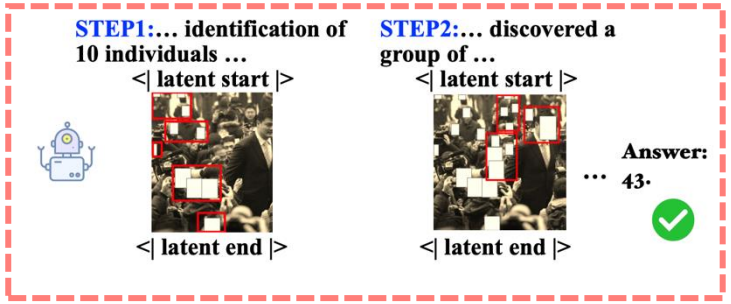


Trace Coherence (Local Plausibility): Do the individual, step-by-step sentences make sense? Are there obvious errors between step A and step B?

Trace Validity (Global Validity): Does the entire chain of thought logically sound that leads to the correct final answer

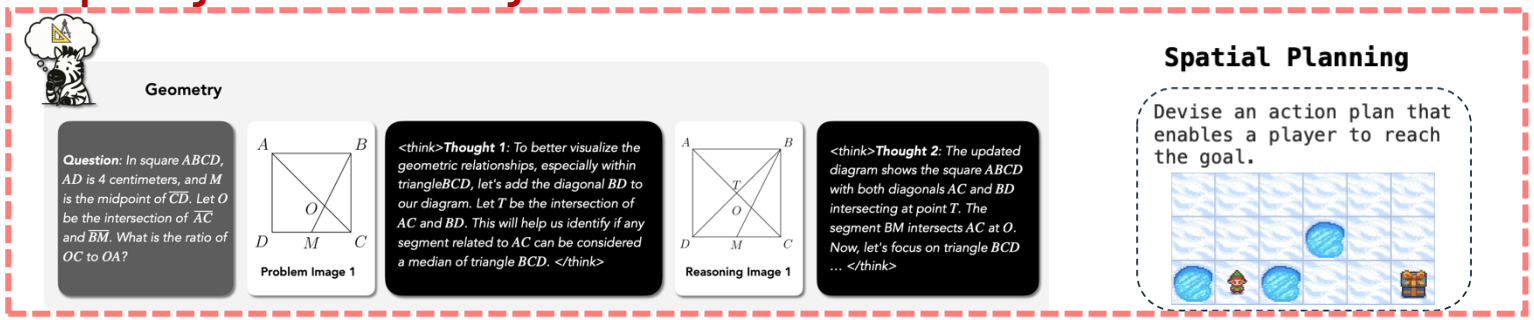
Existing multi-model reasoning data

Highlight task-related region



Caption the input image

2D spatial/geometric reasoning with sketch



3D spatial reasoning with toy data

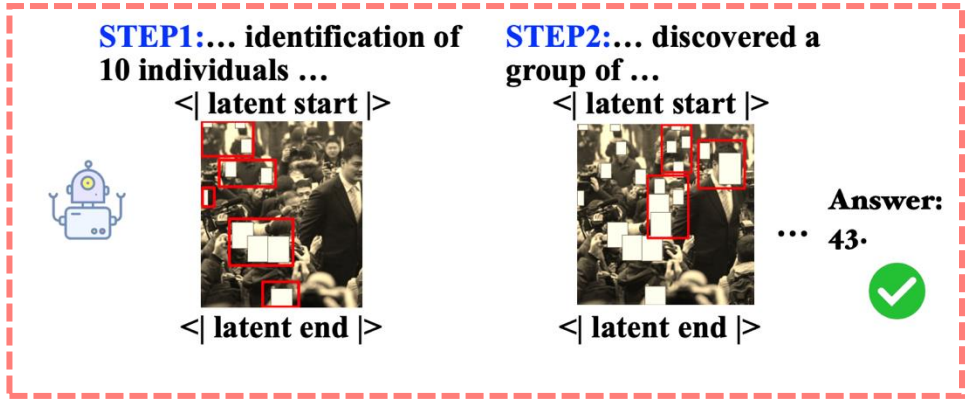
Data Source	Problem Domain	Visual Operation Type	Amount
ReFocus [12]	Chart	Drawing bounding boxes, highlighting	0.4K
CogCoM [24]	Real-world, chart	Cropping, drawing auxiliary lines and bounding boxes	0.5K
Visual-CoT [26]	Real-world, documents, chart	Cropping, drawing bounding boxes	118.6K
Zebra-CoT visual search [18]	Real-world, documents, chart	Cropping, drawing bounding boxes	2.7K
Zebra-CoT geometry [18]	Geometry	Drawing auxiliary lines, creating new geometric sketchpads	0.1 K
Zebra-CoT count [18]	3D object counting	Creating new images with 3D objects removed or added	2.9K

Statistics of monet-sft-125k

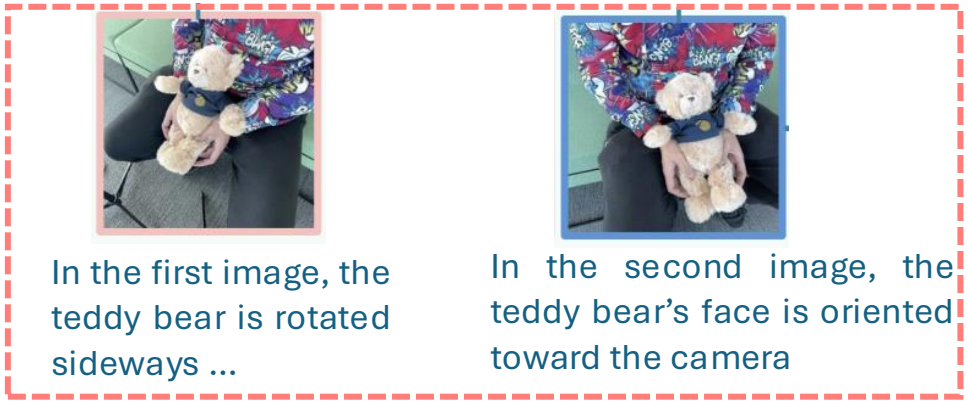
Latent reasoning for spatial reasoning

Existing work: Only highlights task-relevant 2D information, without enabling 3D reasoning.

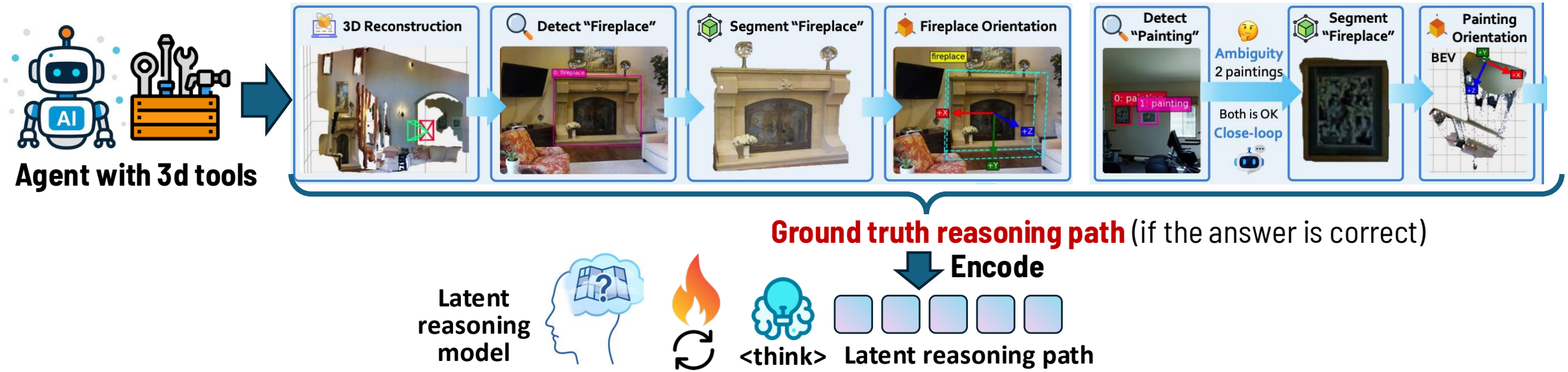
Highlight task-related region



Caption the input image

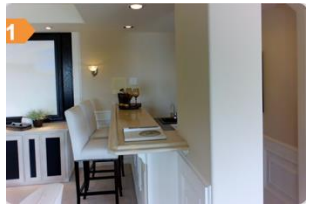


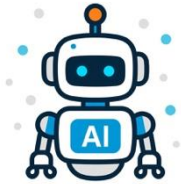
Our idea: Tool calling agents as a teacher

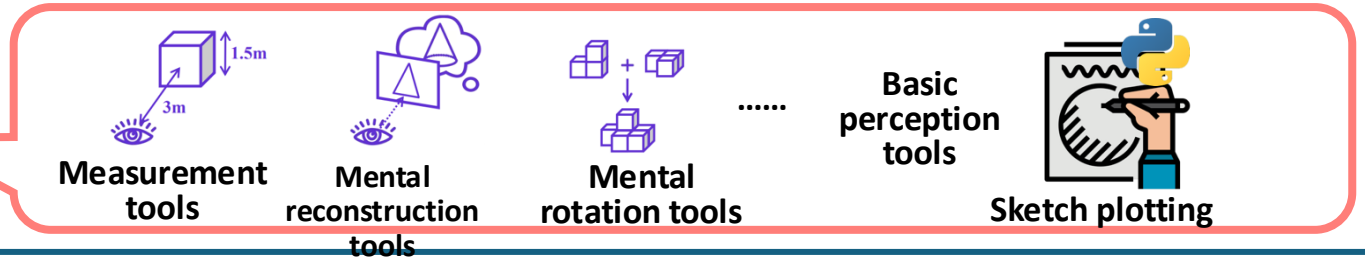


Tool-Calling Agents as Ground Truth for Latent Reasoning Traces

What is the direction of the glass coffee table relative to the book on the bar counter (wine glass is north of the book)?





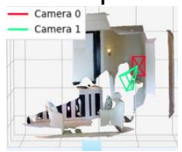


1 The prompt defines "North" as the direction from the book to the wine gla$+Z_{\text{ref}} = \text{Centroid}(\text{glass}) - \text{Centroid}(\text{book})$

2 The target positional relationship is the direction of the coffee table relative to the book. This is represented by the displacement vector $\mathrm{v}_{book \to table}$

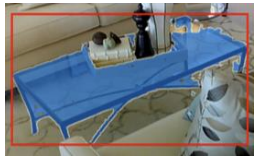
3 I must bridge the 2D visual inputs to 3D spatial data to calculate the defined constraints.

3.1 Reconstruction: I utilize the dual camera views to generate a 3D structural map of the scene. *<3D point cloud of room>*

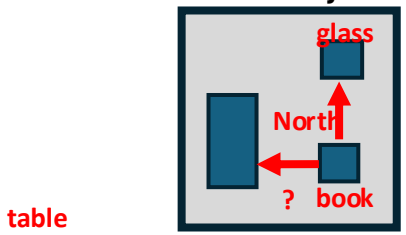


3.2 Detection: I need to locate the target objects within the 2D views to anchor them.
<cropped image: ”table")>
<cropped image: ”book" >
<cropped image: "wine glass" >

3.3 Segmentation and Extraction: Using the 2D detections, I segment the 3D reconstruction to **highlight** the specific 3D point cloud arrays for each target object.
<3D point cloud array: book_points>
<3D point cloud array: wine_glass_points>
<3D point cloud array: coffee_table_points>

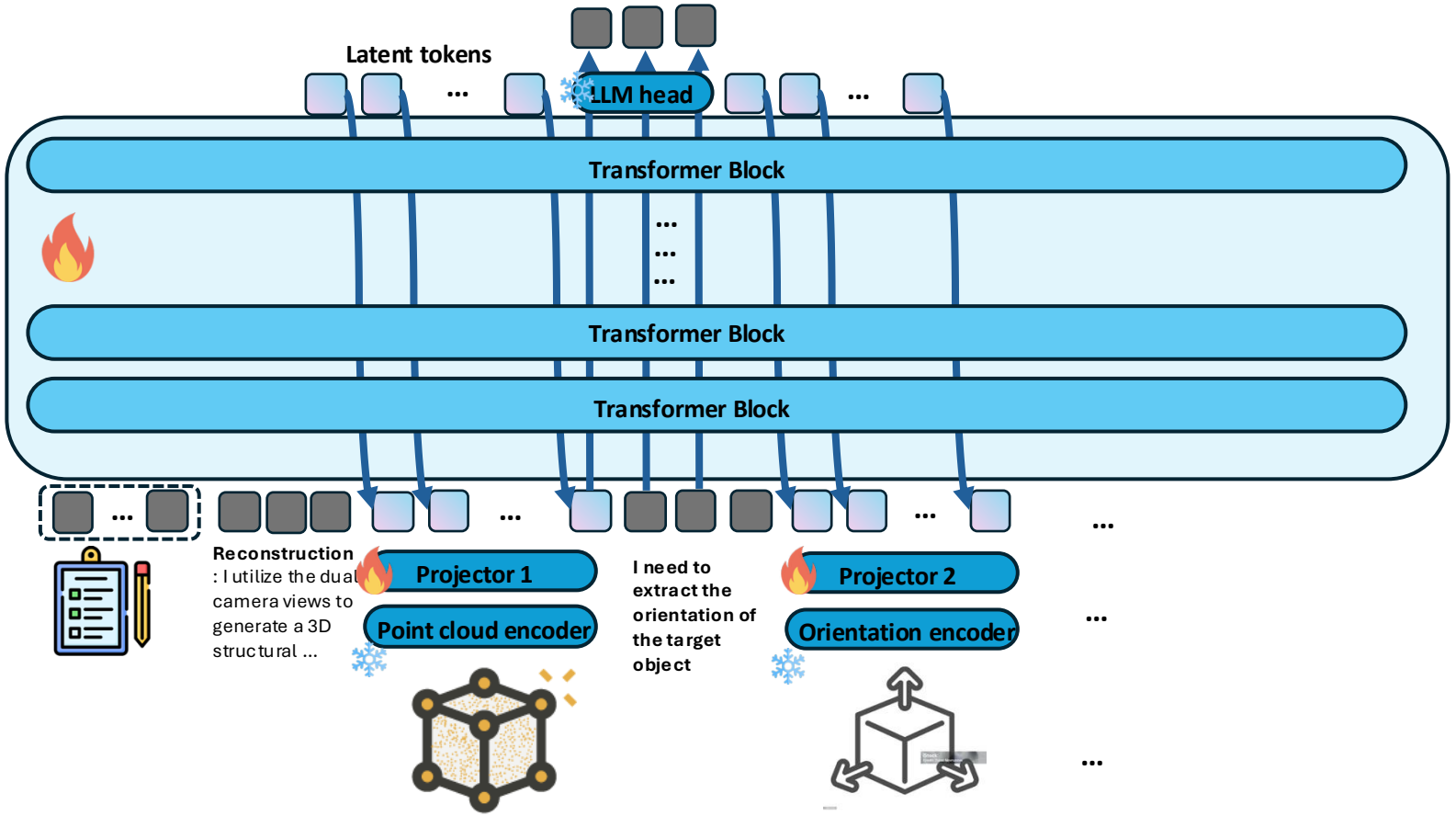


4 Plot sketch to visualize the object location in BEV view



5. Final Conclusion: the direction of the glass coffee table relative to the book on the bar counter is ....

Training steps



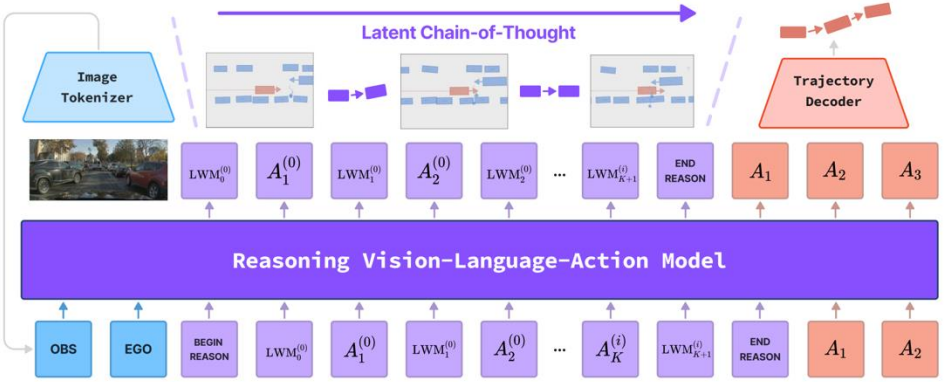
Stage 1: Modality Alignment, train the projectors for LLM to understand multi-model inputs
Stage 2: Latent Thought Grounding, train the model to output **latent multimodal tokens (**supervise these generated latent tokens**)**
Stage 3: End-to-End Reasoning, Optimize the model to use its latent thoughts to arrive at the correct final text answer. (Text-Only Supervision)
Stage 4: Reinforcement Learning, Maximize the accuracy of the multi-step spatial reasoning.

Latent reasoning for spatial reasoning

Three potential pathways

3 world model guided latent reasoning

Advantage: easy to get reasoning trace ground truth (predicted world state based on action)



Limitation: not applicable to all tasks

Applicable: Where is object A relative to me if turn right

Future world state (e.g. image after turning)

Condition (action)

Not applicable: What's the spatial relation between object A and B

What's the action?